

- Exam question
(directly from PP)

$h \times n$

$$n$$

$$\frac{dx}{dt} = \begin{bmatrix} \frac{d}{dt}(\sin t) \\ \frac{d}{dt}(\cos t) \end{bmatrix} = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$$

$$= \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$$

$$\lambda_1, \lambda_2, \dots, \lambda_n$$

... , λ_k

ex) A 3×3

λ_1

λ_2

alg. multiplicities

$\vec{v}_1$

$\vec{v}_2, \vec{v}_3$

geo. mult. 2

L.I.

$\lambda_1 \neq \lambda_2$

- For each eigenvector v^j of A with eigenvalue λ_j , we have a solution $x^j(t) = e^{\lambda_j t} v^j$ of (*). If A has n linearly independent eigenvectors $v^1, \dots, v^n$ with eigenvalues $\lambda_1, \dots, \lambda_n$ respectively ($\lambda_1, \dots, \lambda_n$ need not be distinct), then $x^j(t) = e^{\lambda_j t} v^j$, $j = 1, \dots, n$ are n linearly independent solutions of (*).

Since $\vec{x}(t) = e^{\lambda_j t} \vec{v}_j$ satisfy the equation $\frac{d\vec{x}}{dt} = A\vec{x}$

$$\vec{x}(t) = e^{\lambda_j t} \vec{v}_j \text{ is a solution.}$$

3. Example: Let

$$A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix} \quad 3 \times 3$$

The characteristic polynomial $p(\lambda) = \det(A - \lambda I) = (1 - \lambda)(\lambda - 3)(\lambda + 2)$.

Eigenvalues of A: $\lambda_1 = 1, \lambda_2 = 3, \lambda_3 = 2$

corresponding eigenvectors:

$$v^1 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}, \quad v^2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v^3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

General solution is

Linear $\frac{d\vec{x}}{dt} = A\vec{x}$ Linear

What if A do not
have n
L.I. eigenvectors?

$\frac{d}{dt}(f+g) = f' + g'$
 $\vec{x}^1(t) = e^{2t} \begin{bmatrix} -1 \\ 4 \end{bmatrix}$
 $\frac{d}{dt}(cf) = c f'$

$\vec{x}^2(t) = e^{3t} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$\vec{x}^3(t) = e^{-2t} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

Sum of
geo. mults.
eigenvals
 $\neq n$

$\vec{x}(t) = c_1 \vec{x}^1(t) + c_2 \vec{x}^2(t) + c_3 \vec{x}^3(t)$